

Injectable Nanocarriers for Targeted Drug Delivery in Cancer Therapeutics

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Abstract

This review will focus on the several types of injectable nanocarriers that are currently undergoing immense research but have yet to reach clinical trials for commercial usage (e.g. liposomal-encapsulated doxorubicin). These include DNA based, silica-based and gold nanoparticles, which have shown much benefits in targeted delivery systems administered intravenously, and show great potential as an alternative candidate for future applications given enough time for proper planning and development of these systems to overcome the barriers to delivery. Here, we provide a summary of the challenges that these delivery systems face both in academia and industry, along with methods of improvement and recently published designs that tackle these issues.

Key words: nanocarriers, intravenous, systemic delivery, active targeting

1. Introduction

Cancer has statistically been one of the leading causes of death in humans around the globe, with 8.93 million deaths annually, only second to cardiovascular diseases⁹. Once a person contracts this deadly illness, the person's ten-year survival rate on average is only 52%, despite the large number of treatments commercially available for patients. Nonetheless, most of these treatments, with chemotherapy being most widely used to date, pose many problems or side effects to patients. While these chemotherapeutic drugs are effective in killing tumor cells, the most commonly mentioned problem is that it kills healthy tissues as well, along with its non-specific biodistribution and insolubility due to their hydrophobic nature. This has led to calls for scientists or researchers to come up with new methods for delivery that has the capability to effectively and efficiently target cancer cells, while minimizing other potential issues such as biocompatibility and effects on other vital organs in the body. This is where nanoparticles have shown its massive potential as carriers for in vivo drug delivery. The improved solubility and bioavailability provided by using nanocarriers for chemotherapeutic drugs allows for better accumulation and a facilitated entry of the particles into the target cell, resulting in high specificities that cannot be observed in conventional chemotherapeutic treatments.

The targeting mechanism for intravenously administered nanoparticles is divided into passive and active targeting. Passive targeting works by taking advantage of the leaky vasculature around solid tumors, in which the small size of nanoparticles will be able to pass through these leaky holes and thus accumulate at the tumor site. This effect is known as the Enhanced Permeability and Retention Effect (EPR), which is the core mechanism governing passive targeting of nanoparticles for systemic drug delivery. However, in practice only 0.7% of intravenously administered nanocarriers reaches the target tumor via EPR⁸. While this number may seem small, this delivery efficiency is significantly higher than conventional chemotherapeutic formulations in current clinical applications which is only around 0.2%. This

proves the immense benefits from utilizing nano-based drug delivery systems for maximum therapeutic effectiveness of drugs to the tumor site. Nonetheless, there is much research going on in this field to further improve the delivery efficiency of these systems, many in which are still undergoing preclinical trials. Moving on, active targeting strategies involve the incorporation of affinity ligands being attached to the surface of the nanocarriers for specific honing, increasing the retention at the target tumor along with its uptake by the targeted cells. These ligands are often engineered to selectively bind to overexpressed receptors on the diseased cells.

There are several targeted delivery methods for nanoparticles to be administered in vivo for in vivo applications. These include injection, inhalation, oral, or even the use of hydrogels and microdevices for the delivery of these nanotherapeutics. In this review we will focus on the injection of the types of nanocarriers via the intravenous route, and we will address on the many challenges that arise when designing these systems. While this review does not discuss much on the commercially available liposomal-encapsulated doxorubicin (Doxil) or polymeric nanocarriers, we focus our attention on less commonly used delivery systems that has recently made major research advances and show great potential as an alternative for future clinical translation.

2. Challenges & Potential Ways to Improve Delivery Methods

While there are some passively targeted nanocarriers currently available in the market, it is unfortunate that there has yet been any actively targeting nanocarriers which has progressed past clinical trials despite its massive potential in cancer therapeutics. This section discusses the major challenges/barriers that stand in the way of these delivery systems, along with the potential ways to improve their success for future clinical applications. To start off, the challenges faced by the delivery of nanocarriers is broken down into 2 stages, accumulation and retention along with penetration and post-penetration.

Accumulation & Retention (At Tumor Site)

In the systemic delivery of nanocarriers, intravenously administered nanoparticles often heavily rely on the Enhanced Permeability Effect (EPR), the main mechanism for passively targeting nano-systems to accumulate at the tumor site. This method of delivery though beneficial, poses many problems that comes with the nature of systemically delivered therapeutics. One of these is the clearance of nanocarriers from the body (e.g. by mononuclear phagocytic system (MPS)), which causes a significant reduction in the half life and therapeutic efficacy of the drug⁸. Besides that, as only an extremely small percentage (<1%) of the nanocarriers actually end up at the tumor site from EPR, a major portion of the nanoparticles delivered in vivo accumulates in other organs such as the heart and liver, resulting in toxicity issues. Unfortunately, the toxicity of these chemotherapeutics in vital organs is still a big hurdle to pass in clinical trials. Finally, pre-mature release of therapeutics from the nanocarrier before it internalizes in the target cell also reduces its overall therapeutic efficiency, a problem that has shown much progress in recent years^{4,6}. Due to the inefficiency of EPR in delivering cancer nanotherapeutics, one of the methods that is used to

complement this involves utilizing the natural tropism of certain types of cells towards the targeted cells, which will be further discussed in the next section¹⁰.

In active targeting, the purpose of the design in these systems is to increase the retention and hence the localization of nanocarriers at the tumor site, which is done by conjugating targeting ligands on the surfaces of nanocarriers^{3,11}. One of the main issues associated with this is that they do not consider the heterogeneity of tumor cells. This is because current active targeting nanocarriers are designed to only focus on a single receptor on the surface of the tumor cell, resulting in potential resistant tumor relapses⁸. A possible way to overcome this would be to combine multiple types of nanocarriers in delivery systems, though that itself may bring a whole new level of complexity in the manufacturing of these nanomedicines. Nonetheless for systemic delivery of these nanocarriers, it is crucial that efficient passive targeting is first achieved before any means of active targeting, as the nanoparticles must first be able to reach the target cell before utilizing the full benefit of its increased affinity and avidity.

Penetration & Post-Penetration

Following localization of the nanocarriers at the tumor site, physiological barriers prove to be one of the biggest challenges for its penetration into the tumor cell. Understanding of physicochemical properties such as the size, shape, pH or surface effects of nanoparticles is vital in optimizing the design of nanocarriers. Several methods have been conducted to overcome these barriers such the incorporation of cell-penetrating peptides (CPPs) at the surface of the nanoparticles, though they tend to have non-specific targeting, a short half-life, and severe toxicity for in vivo applications¹¹. To tackle this, Zhang, Jiang, Zeng et al. took advantage of the chemical conditions of tumor cells to mask a polymeric nanocarrier with acidity-activatable arginine-rich (R9) peptide, a type of CPP which enhances cell penetration¹¹. As the name suggest, the shielded R9 peptide nanocarrier is able to effectively localize at the tumor site, in which the slightly acidic (pH 6.5) tumor extracellular environment will then de-shield the acid from the nanoparticle for its penetration effects, which translates to a higher tumor cell uptake of the drug.

The final challenge that is present once the nanocarrier has successfully penetrate and internalize in the tumor cell is its escaping from the endocytic pathway in vivo. As supposedly in an ideal scenario the nanocarrier would work its way in the cell by intracellular tracking mechanisms, degradation of the nanocarrier can occur before it can reach the perinuclear region of the target cell for maximum therapeutic effect. One crucial step in approaching this would be to first understand the degradation mechanism and its criteria, and then incorporate that knowledge in the design of nano-delivery systems to improve the efficiency of drug delivery systems.

To end on a note, it appears to be extremely common that supposedly groundbreaking discoveries in these nano-based delivery systems have such low rate of clinical translation despite its massive potential. The main challenge here especially in the pharmaceutical sector is the transition of bridge from academia to industry, as many researchers tend to publish results that are not reproduceable or unscalable/non-practical for industry purposes. It is essential that

the scientific community and industry leaders in this sector strive to better facilitate this gap, which would include substantial effort and collaboration from both ends. These include good lab practices from researchers and essential knowledge on the manufacturing process of these drug delivery systems, as it would not be fruitful for patients in the real world if a major breakthrough in the cure for cancer for instance only producible in a lab setting.

3. Injectable Nanocarriers in Systemic Drug Delivery

DNA Nanostructures as drug-delivery Nanovehicles (DNPs)

DNA based nanoparticles, or DNP in short have also seen its applications for in vivo drug delivery systems. These DNA nanostructures (50-400 nm) are often developed as carriers for siRNA or chemotherapeutic drugs, hence it is compatible for utilization of the EPR effect for systemic delivery. One of the main benefits in DNA nanotechnology is that DNPs can be produced with great control of the size and surface functionality, which allows for easier preparation and better targeting of nanocarriers for highly specific cancer treatments^{4,7}. In addition to that, the ability to readily modify structure for a stimuli-triggered configuration allows for a controlled drug release in tumor cell, showing potential in the development of intelligent drug delivery systems.

In a recent research, Rahman, Wang, Zhao et al. have shown the use of DNPs to provide for an efficient nano-delivery system for siRNA, which is used for targeted gene silencing in tumor cells⁷. Here, siBcl2 is used to target the overexpressed Bcl-2 gene, which is common in some cancers such as lung and prostate cancer as the gene contributes to the development of tumor and resistance to conventional therapies. An experiment conducted on intravenously delivered DNPs in mice demonstrated the use of siBcl2 to silence Bcl2, inhibiting its expression in tumor cells, hence suppressing tumor growth. While there was evidence of toxicity in the mice, one of the main problems with DNPs is that its active targeting is relatively weak compared to other nanocarriers as mentioned above, which may cause a problem for other healthy tissues. This is due to DNPs entering cancer cell via scavenger receptors, which may cause unwanted immune responses. A possible solution for this would be to modify or conjugate DNPs with targeting ligands, hence enhancing the benefits of active targeting for a more efficient uptake in cancer cells. Finally, DNPs by itself are often unstable for long term therapeutic treatment, due to its fast degradation profiles in vivo. Despite this, various methods have been used to overcome this, such as to bind block copolymers with DNPs to form hybrids via electrostatic interactions to protect it from nuclease digestion, thus increasing its half-life and overall stability for intravenously administered delivery systems¹.

As a conclusion, DNA nanostructures, if engineered properly can be made into highly specific and practical systems as an alternative candidate for the deliveries of drugs or genes for cancer treatment. The large freedom to design and program DNA nanostructures in varying shapes and sizes meant that it has a comparable advantage to other delivery systems including polymeric or liposomal carriers, which tends to be limited in that regard⁴. Nonetheless, several other critical challenges with DNPs must first be addressed before it can show its potential for future clinical translations, which include its targeting effectiveness, possible immune responses and clearance mechanisms for in vivo intravenous administrations.

Silicon or Silica-Based Nanocarriers

Delivery systems involving silica-based nanocarriers have received much attention as an alternative due to its large surface area and advantageous chemical properties which allow for a diverse set of modifications to enhance cell uptake and better control drug release. As discussed above, one of the major hurdles with nanocarriers in drug delivery systems is the presence of multiple physiological barriers. In this subsection, the potential of a silicon-based injectable nanoparticle generator (iNPG) created by Xu, Zhang, Mai et al. which bypasses these issues is further explored, as this discovery has indeed a great promise for clinical translations¹⁰.

The design of this nanocarrier first starts of by conjugating Doxorubicin (DOX) to polylactic-co-glycolic-acid (PLGA) polymer with a pH-sensitive cleavable linker, forming pDox in short. These pDox monomers are then loaded into the nanopores of a micrometer size discoidal silicon-based carrier (iNPG), in which the packing is designed to optimize the concentration of nanoparticles at the tumor site. When administered intravenously, the main benefit of using the iNPG is because it localizes and accumulates extremely well at the tumor site due to natural tropism & favorable vascular dynamics, thus minimizing accumulation in other vital organs as tested in mice. Once the iNPG accumulates at the tumor from leaky vasculatures, the pDox monomers will then self-assemble into nanoparticles from exposure to an aqueous environment, which are then released in a sustained manner from the iNPG (Figure 1). Following internalization in tumor cells, pDox then undergoes trafficking to perinuclear regions of the tumor cell, where the acidic environment of endosomes cleaves the pH-sensitive linker on the nanoparticles to release high concentrations of DOX directly at the nucleus.

Besides that, it was also demonstrated that intraperitoneal administration of the iNPG on mice resulted in lower drug levels at the tumor site compared to intravenous administration. This shows that nanocarriers containing chemotherapeutic drugs are still most effective when delivered directly into the bloodstream for solid tumors, although the heavy reliance on EPR effect may be ineffective in some types of cancers such as leukemia. Nonetheless, one of the downfalls of this system is that it heavily relies on pH, as slightly acidic environments in the vasculature may cause the pH-dependent cleavage of pDox to overtake nanoparticle assembly, thus rendering the use of the iNPG system unbeneficial. To conclude, results have shown that this silicon-based delivery system to be more effective than nanoparticle-encapsulated DOX or commercially available Doxil (liposome-encapsulated DOX), thus making it an excellent candidate for further clinical trials.

In another study by Liu, Li, Zhang et al., mesoporous silica-based nanoparticles (MSNs) were also used to overcome the barrier of drug resistances at tumor cell membranes⁶. Here, the MSNs were coated with sericin, which not only bypasses these barriers causing drug efflux, but also prevents premature leaking of the encapsulated DOX drug. Following internalization, the acidic environment of perinuclear lysosomes breaks down the pH sensitive sericin coating, leading to a burst of DOX in the tumor cell (Figure 2).

All in all, silica or silicon-based drug delivery systems has demonstrated its ability to overcome many problems present in current systems in clinical applications, which include multiple drug resistances and toxicity. It is observed that a common strategy was being incorporated in these delivery systems which takes advantage of physiological conditions at the tumor cell when designing these nanocarriers, where pH levels were the most common stimuli to trigger release from these nanocarriers. It is interesting to note that the high surface area and biocompatibility of silica based nanocarriers in vivo, along with its ability for a diverse set of modifications prove that they can be potentially utilized as an effective drug delivery system for clinical translation.

Gold Nanoparticles

Gold nanoparticles (AuNPs) have also played its role in many applications throughout cancer treatment including imaging capabilities for diagnostics or for drug delivery purposes. One of the main benefits of AuNPs as nanocarriers for chemotherapeutic drugs is its surface chemistries allowing for an easily functionalized surface and numerous types of molecules that can be attached to its surface³. An example of this would be the synthesis of a PEGylated gold nanoparticle conjugate, which allows for a hydrophobic drug to be soluble in water and increased delivery efficiency due to its enhanced targeting features².

Nonetheless one of the main problems associated with naked AuNPs is its extreme toxicity. However, with proper surface modifications such as in the example above the toxic effects from pure gold nanoparticles can be reduced significantly or even eliminated for in vivo applications. All in all, there is still much to be understood in the mechanism of AuNPs for systemic targeted delivery of chemotherapeutics, as researchers must first seek this before hopes of any AuNPs systems being transitioned for clinical trials. Albeit challenging, it is positive that AuNPs will be able to achieve its potential in this field of cancer nanotherapeutics given enough time and proper planning for future commercial usage.

4. Conclusions & future prospects:

To sum up, these nanocarriers while relatively new, have shown much potential in systemic drug delivery for cancer therapeutics due to their ability to overcome the many challenges or barriers that arise when designing these systems in vivo. The pros and cons for each type of nanocarrier were summarized in Table 1 below. In addition to this, it was observed that much effort has been done through the integration of multiple types of nano-delivery systems in cancer therapeutics for their combined benefits and complementing each other's downfalls^{1,5,10}. While this may substantially raise the complexity of designs and may result in additional problems, this is the ultimate goal in designing of delivery systems, especially with the many natural limitations/challenges that come when dealing with biological systems. One good example of this as mentioned earlier would be the utilizing block copolymer micellization as a protective coating for DNA nanoparticles¹. Here, the polymeric micelle formation protects DNPs from its

ease in degradation, while still having the high specificity benefits of DNPs as a drug delivery system. Nonetheless, this obviously requires the maturing of each fields/types of nanocarriers before the full potential of cross-applications can be realized when designing them.

Finally, as most of the drug delivery systems have massive potential but are still in preclinical modeling/trials, we still have a long way to go before it is commercially available in the market⁸. It is indeed a frustrating fact, but it is the nature of this field when only an extremely small percentage of the drugs in preclinical trials make its way to being widely approved, a process which takes many years to accomplish. Nonetheless, many scientists opt not to take the additional effort or step to ensure this, as it may cost them additional time and money for the approval process (regulations etc.) which they would rather invest it on further research. In the meantime, I do hope that researchers would continually strive to incorporate the upscaling of the drug delivery systems in mind when designing such systems. This can be accomplished through effective collaboration between experts from different sectors such as engineers who have sufficient knowledge on this field. Through this practice, it is certain that there would a higher success rate and potential for a drug delivery system to be widely produced for those who need it, which is the ultimate dream for any researcher in this aspiring field.

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Figures & References

Types of Injectable Nanocarriers	Advantages	Disadvantages
Silicon / Silica	* Large surface area-volume ratio (high drug loading) * Stable and biocompatible * Ease of structure modifications	* Slow biodegradation
Gold	* Highly functionalizable surface * High Drug Loading	* Toxic
DNA	* Ease of preparation * Highly specific (Ease of surface modifications) * High control / freedom in designing systems	* Low stability (Degradation) * Weak active targeting * Possible immune response

Table 1

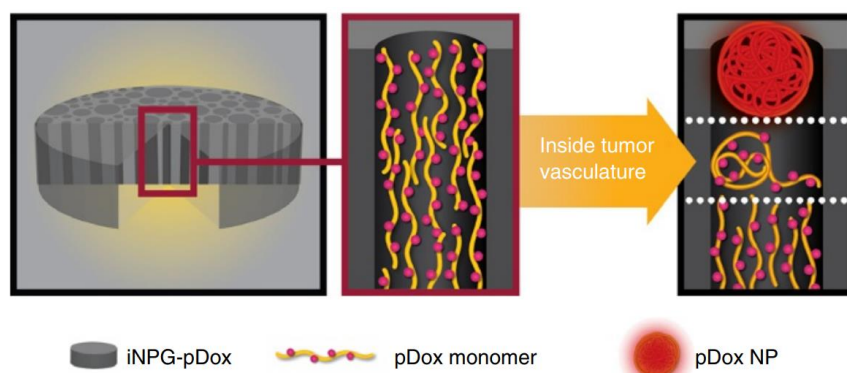


Figure 1. 1 iNPG-pDox characterization and pDox assembly and release from iNPG. Schematic diagram depicting iNPG-pDox composition, pDox prodrug encapsulation, and pDox nanoparticle assembly and release from nanopores.

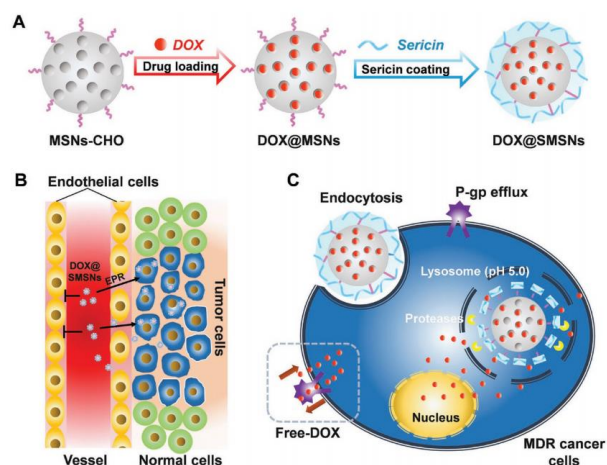


Figure 2: Schematic illustration of sericin-coated MSNs (SMSNs) preparation and their delivery route from blood circulation to lysosomes of tumor cells. A) The synthesis procedure of DOX-loaded sericin-coated MSNs (DOX@SMSNs). B) DOX@SMSNs enter tumor tissue from blood circulation through EPR effects. C) Cellular uptake of DOX@SMSNs and pH/protease triggered lysosomal drug release from SMSNs in multidrug resistant (MDR) cancer cells.

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